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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Author(s)           | Zixian Gong (Works count: 2, Citations count: 0, H-index: 0), Kun Tian (Works count: 2, Citations count: 0, H-index: 0), <b>Yi Zhang</b> (Works count: 115, Citations count: 1630, H-index: 22), Fengxia Liu (Works count: 2, Citations count: 0, H-index: 0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Item Type           | Article                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Abstract            | Threshold private set intersection (TPSI) allows parties to reveal their intersection only when its cardinality reaches a prescribed threshold. Existing quantum TPSI protocols typically rely on a third party (TP) to interpret the final results, which deviates from the cardinality-testing paradigm of TPSI. In this paper, we propose a quantum multiparty TPSI protocol with explicit cardinality testing. Our protocol develops a rotation-based quantum construction in which single-photon sequences are sequentially processed through participant-side data rotations, TP-participant masking rotations, and correlated aggregate rotations. <b>This design produces hidden-label measurement vectors: TP can complete the final measurement, but cannot interpret the semantic meaning of the outcomes.</b> Based on these hidden measurements, we further realize the threshold decision through an oblivious linear evaluation (OLE)-based inner product procedure and a lightweight garbled circuit, revealing only $\mathbf{1}[\bigcap_i X_i \geq \tau]$ before conditional intersection reconstruction. We prove the correctness and security of the proposed protocol, and further validate its feasibility through quantum-circuit simulations implemented on the IBM Qiskit platform. |
| Date                | 2026-06-26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Citation Key        | GONGQuantumMultiPartyThresholdPrivateSetIntersection2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DOI                 | <a href="https://doi.org/10.48550/arXiv.2606.27996">10.48550/arXiv.2606.27996</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| URL                 | <a href="http://arxiv.org/abs/2606.27996">http://arxiv.org/abs/2606.27996</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Importance          | New / Niche ( $\geq 10$ citations)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| FWCI                | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Citation percentile | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Cites               | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cited by            | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Related to          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Retracted           | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| PubPeer Alerts      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| OpenReview Alerts   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Peer Reviewed       | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### 3. Agentic AI-Powered Re-Identification: An Emerging, Scalable Threat to Mobility Microdata Privacy

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| Author(s) | Oscar Thees (Works count: 6, Citations count: 5, H-index: 2), Roman Müller (Works count: 4, Citations count: 0, H-index: 0), Matthias Templ (Works count: 213, Citations count: 4755, H-index: 31) |
| Item Type | Article                                                                                                                                                                                            |